YUNSONG ZHOU

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EDUCATION BACKGROUND

Shanghai Jiao Tong University, Ph.D., School of Computer Science *2020.09 - 2025.09*
Advised by Prof. Minyi Guo and Prof. Hongzi Zhu
Wu Wenjun AI Honors Class, head teacher: Prof. Cewu Lu
University of California, Los Angeles, Department of Computer Science *2024.01 - 2024.07*
Visiting scholar (Onsite), advised by Prof. Bolei Zhou
Shanghai Jiao Tong University, B.S., Microelectronics Science and Engineering *2016.09 - 2020.06*
School of Electronic Information and Electrical Engineering

WORK EXPERIENCE

Shanghai AI Lab, Research Scientist, leading dexterous manipulation *2025.06 - present*
Shanghai AI Lab, Research Intern in generative models and autonomous driving *2022.12 - 2025.06*
SenseTime, Research Intern in Mobile Agent Perception group *2020.03 - 2020.12*

AREAS OF INTERESTS

Generative models, robotics (dexterous manipulation), and embodied AI.

HIGHLIGHTS

Area Chair, ICLR 2027 and ICML 2027 *2026*
SAC Highlight Award, EMNLP 2026 (nominated to the Best Paper Committee) *2026*

SELECTED PUBLICATIONS

[1] **Yunsong Zhou**[*,†], Xuekun Jiang[*], Haiqi Shen[*], Hangxu Liu, Chaojian Shi, Yujie Liu, Jingyuan Li, Haoran Yang, Jiafei Cao, Dongzhan Zhou, Dong Wang, Yao Mu, Chunhua Shen, Weinan Zhang[†]. Ego-Explorer: A Vision-Motion-Touch Observatory for Human Loco-Manipulation In Situ. *arXiv preprint*. 2026.

[2] **Yunsong Zhou**[*,†], Hangxu Liu[*], Xuekun Jiang[*], Xing Shen[*], Yuanzhen Zhou, Hui Wang, Baole Fang, Yang Tian, Mulin Yu, Qiaojun Yu, Li Ma, Hengjie Li, Hanqing Wang, Jia Zeng, Jiangmiao Pang[†]. SIM1: Physics-Aligned Simulator as Zero-Shot Data Scaler in Deformable Worlds. In *ECCV*. 2026.

[3] **Yunsong Zhou**, Quan Liu, Hongzi Zhu, Yunzhe Li, Shan Chang, and Minyi Guo. Exploiting Ground Depth Estimation for Mobile Monocular 3D Object Detection. In *TPAMI*. 2025. (Conference version MoGDE: NeurIPS 2022, **Spotlight**.)

[4] **Yunsong Zhou**, Yuan He, Hongzi Zhu, Cheng Wang, Hongyang Li, and Qinhong Jiang. MonoEF: Extrinsic Parameter Free Monocular 3D Object Detection. In *TPAMI*. 2021.

PUBLICATIONS AND PREPRINTS

[1] Chaojian Shi, Renhong Huang, Xulan Qiu, Jiawei Zhou, **Yunsong Zhou**. Modular Representation Steering for Multi-Attribute LLM Alignment. In *EMNLP*. 2026. **SAC Highlight Award; Best Paper Committee nominee.**

[2] Kaipeng Fang, Hangxu Liu, Chaojian Shi, **Yunsong Zhou**[†], Hongzi Zhu, Minyi Guo, Jingkuan Song, and Lianli Gao[†]. Data-Free Continual Learning for World Action Model with Rectified Generative Replay. *arXiv preprint*. 2026.

[3] Xingyu Miao, Zizun Li, Baole Fang, Kaiwen Song, …, **Yunsong Zhou**, …, Tianfan Xue, Weinan Zhang, and Chunhua Shen. InternW0-Δ: A World Action Model Bridging Predictive Dynamics and Actions with 20K+ Hours of Open Data. *arXiv preprint*. 2026.

[4] Shuai Wang, Yaxin Feng, Xuekun Jiang, Shihan Tian, Ningyu Yan, Xing Shen, Chaoyang Lyu, Hui Wang, **Yunsong Zhou**, Hanqing Wang, Jiangmiao Pang, Yang Xiang, Xing Gao, Chunhua Shen, and

Weinan Zhang. GAUGE: A Measurement-Grounded Benchmark for Physical Fidelity in Simulation Engines and Video World Models. *arXiv preprint*. 2026.

[5] Mu Huang, Hui Wang, Kerui Ren, Linning Xu, **Yunsong Zhou**, Mulin Yu, Bo Dai, and Jiangmiao Pang. SoMA: A Real-to-Sim Neural Simulator for Robotic Soft-body Manipulation. In *ICML*. 2026.

[6] Yang Li, Hongru Jiang, Junjie Xia, Hongquan Zhang, Jinda Du, **Yunsong Zhou**†, Jia Zeng†, Ce Hao, Jieji Ren, Qiaojun Yu†, Cewu Lu†, Yu Qiao, Jiangmiao Pang. ForceVLA2: Unleashing Hybrid Force-Position Control with Force Awareness for Contact-Rich Manipulation. In *CVPR*. 2026.

[7] Junhao Cai, Zetao Cai, Jiafei Cao, Yilun Chen, …, **Yunsong Zhou**, Hongrui Zhu, Yangkun Zhu, and Yuchen Zhu. InternVLA-A1: Unifying Understanding, Generation and Action for Robotic Manipulation. *Arxiv preprint*.

[8] **Yunsong Zhou***, Naisheng Ye*, William Ljungbergh, Tianyu Li, Jiazhi Yang, Zetong Yang, Hongzi Zhu, Christoffer Petersson, and Hongyang Li. Decoupled Diffusion Sparks Adaptive Scene Generation. In *ICCV*. 2025.

[9] **Yunsong Zhou**, Micheal Simon, Zhenghao Peng, Sicheng Mo, Hongzi Zhu, Minyi Guo, and Bolei Zhou. SimGen: Simulator-conditioned Driving Scene Generation. In *NeurIPS*. 2024.

[10] **Yunsong Zhou***, Linyan Huang*, Qingwen Bu*, Jia Zeng, Tianyu Li, Hang Qiu, Hongzi Zhu, Minyi Guo, Yu Qiao, and Hongyang Li. Embodied Understanding of Driving Scenarios. In *ECCV*. 2024.

[11] Quan Liu, Hongzi Zhu, Zhenxi Wang, **Yunsong Zhou**, Shan Chang, and Minyi Guo. Extend Your Own Correspondences: Unsupervised Distant Point Cloud Registration by Progressive Distance Extension. In *CVPR*. 2024.

[12] Quan Liu, Hongzi Zhu, **Yunsong Zhou**, Hongyang Li, Shan Chang, and Minyi Guo. Density-invariant Features for Distant Point Cloud Registration. In *ICCV*. 2023.

[13] Quan Liu, **Yunsong Zhou**, Hongzi Zhu, Shan Chang, and Minyi Guo. APR: Online Distant Point Cloud Registration Through Aggregated Point Cloud Reconstruction. In *IJCAI*. 2023.

[14] **Yunsong Zhou**, Hongzi Zhu, Quan Liu, Shan Chang, and Minyi Guo. MonoATT: Online Monocular 3D Object Detection with Adaptive Token Transformer. In *CVPR*. 2023.

[15] **Yunsong Zhou**, Quan Liu, Hongzi Zhu, Yunzhe Li, Shan Chang, and Minyi Guo. MoGDE: Boosting Mobile Monocular 3D Object Detection with Ground Depth Estimation. In *NeurIPS*. 2022. **Spotlight.**

[16] **Yunsong Zhou**, Hongzi Zhu, Chunqin Li, Tiankai Cui, Shan Chang, and Minyi Guo. TempNet: Online Semantic Segmentation on Large-Scale Point Cloud Series. In *ICCV*. 2021.

[17] **Yunsong Zhou**, Yuan He, Hongzi Zhu, Cheng Wang, Hongyang Li, and Qinhong Jiang. Monocular 3D Object Detection: An Extrinsic Parameter Free Approach. In *CVPR*. 2021.

RESEARCH AND WORK EXPERIENCE

Research Scientist, Shanghai AI Lab Supervisors: **Weinan Zhang, Chunhua Shen**

EgoExplorer: A Vision-Motion-Touch Observatory for Human Loco-Manipulation In Situ

- EgoExplorer is a 100-hour vision-motion-touch dataset of human loco-manipulation from a portable wearable system, aligning egocentric video with whole-body motion, finger articulation, touch, and step-level descriptions. Its Calibrate-Complete-Constrain pipeline recovers full-body motion from sparse sensors, cutting body-point error by 71.9%.

R^2-WAM: Data-Free Continual Learning for World Action Model with Rectified Generative Replay

- R^2-WAM lets a world action model learn new tasks in sequence without old data: it regenerates rollouts of earlier tasks, truncates them at task completion and filters them for physical consistency before replay, reducing forgetting by over 90% versus sequential fine-tuning and nearly matching replay with real data.

SIM1: Physics-Aligned Simulator as Zero-Shot Data Scaler in Deformable Worlds

- SIM1 presents a physics-aligned simulated world designed to bridge the sim-to-real gap in deformable environments. It acts as a zero-shot data scaler, generating high-quality training data without requiring task-specific real-world collection. This enables more robust and generalizable policy learning for complex deformable manipulation tasks.

ForceVLA2: Unleashing Hybrid Force-Position Control with Force Awareness for Contact-Rich Manipulation

- ForceVLA2 extends vision-language-action models by incorporating force-aware representations to handle contact-rich manipulation better. It aligns visual, linguistic, and force signals within a unified framework, enabling more precise and robust control during interaction. This leads to improved performance on tasks that require fine-grained physical feedback, such as insertion and compliant manipulation.

SoMA: A Real-to-Sim Neural Simulator for Robotic Soft-body Manipulation

- SoMA introduces a real-to-sim neural world tailored for soft-body manipulation, learning deformable dynamics directly from real-world data. It captures complex, non-linear material behaviors that are difficult to model analytically, enabling more faithful simulation of soft objects. This improves sim-to-real transfer and supports more reliable policy learning for soft-body manipulation tasks.

InternVLA-A1: Unifying Understanding, Generation and Action for Robotic Manipulation

- InternVLA-A1 unifies perception, generation, and action within a single vision-language-action framework for robotic manipulation. It leverages large-scale multimodal pretraining to jointly model scene understanding, task planning, and control. This integration enables more generalizable and flexible manipulation across diverse tasks and environments.

Research Intern, Shanghai AI Lab (InternRobotics) Supervisors: **Jiangmiao Pang**
InternManip: All-in-one Policy Suite for Robot Manipulation

- Developed InternManip, a unified framework supporting training and evaluation of diverse policy models across datasets and benchmarks for general-purpose manipulation. InternManip is selected as the official platform for **IROS 2025 Challenge** of Vision-Language Manipulation in Open Tabletop Environments.

Research Intern, Shanghai AI Lab (OpenDriveLab) Supervisors: **Hongyang Li**
WorldEngine: Towards the Era of Post-Training for Physical AI

- WorldEngine introduces a post-training infrastructure for physical AI that targets long-tail, safety-critical scenarios in autonomous driving. It automatically discovers failure cases from real-world logs, reconstructs them into photorealistic interactive simulations, and expands them via behavior-driven generation. This enables scalable RL-based post-training, significantly improving closed-loop performance without relying solely on pretraining data.

Nexus: Decoupled Diffusion Sparks Adaptive Scene Generation

- Nexus is a noise-decoupled generative framework designed for adaptive and goal-driven driving scene generation. It enables both real-time reactivity and long-horizon control, overcoming the limitations of full-sequence denoising and next-frame prediction. Nexus achieves a 40% reduction in displacement error and improves closed-loop planning performance by 20%, demonstrating effectiveness in reactive scene generation and data augmentation for autonomous driving.

ELM: Embodied Understanding of Driving Scenarios

- ELM is the first embodied language model for understanding the long-horizon driving scenarios in space and time. ELM expands a wide spectrum of new tasks to fully leverage the capability of large language models in an embodiment setting and achieves significant improvements in various applications. This interpretable driving model, on the basis of language prompting, is a main track in the CVPR 2024 Autonomous Driving Challenge.

OpenScene: the largest 3D occupancy prediction benchmark

- OpenScene provides long-horizon, multi-city data with occupancy and motion annotations, supporting unified scene representation beyond traditional detection or segmentation. This facilitates scalable learning of perception and world models for end-to-end driving systems.

Visiting Scholar, University of California, Los Angeles Supervisors: **Bolei Zhou**
SimGen: Simulator-conditioned Driving Scene Generation

- SimGen is the first simulator-conditioned generative model for controllable driving scene generation with appearance and layout diversity. It addresses the simulation-to-reality (Sim2Real) gaps via a cascade diffusion paradigm, and follows layout guidance from simulators and cues of the rich text prompts to realistic driving scenarios. The corresponding DIVA dataset comprises 147.5 hours of web videos and synthesized data for diverse scene generation and advancing Sim2Real research.

Ph.D. Student, Lab of Innovation on Networking Supervisors: **Hongzi Zhu, Minyi Guo**
Perception in Autonomous Driving

- MonoATT: Adaptive Token Transformer for Monocular 3D Detection. Proposed an online Mono3D framework with adaptive tokenization to replace fixed grid-based vision tokens, enhancing mobile detection accuracy and efficiency.
- MoGDE: Depth-Guided Monocular 3D Detection (NeurIPS 2022 **Spotlight**; extended to TPAMI 2025). Introduced a depth estimation module to address near-far disparity, improving detection of distant objects by leveraging estimated ground geometry.
- TempNet: Temporal Fusion for Point Cloud Segmentation. Designed a lightweight segmentation network with multi-frame fusion, significantly improving stability and accuracy on sequential point cloud data.

Research Intern, SenseTime Supervisors: **Hongyang Li, Jianping Shi**
Mono3D: An Extrinsic Parameter Free Approach

- Existing methods on open datasets neglect the camera pose information, which inevitably results in the detector being susceptible to camera extrinsic parameters. MonoEF proposes a novel method to capture camera pose to formulate the detector free from extrinsic perturbation. By doing so, the Mono3D detector works independently of the extrinsic parameter variations and produces accurate results in realistic cases.

HONORS AND AWARDS

Outstanding Ph.D. Graduates of Shanghai Jiao Tong University	*2025*
Shanghai Youth Science and Technology Hope Star	*2022*
Tencent Scholarship	*2021*
Outstanding Graduates of Shanghai Jiao Tong University	*2020*
National Scholarships for Undergraduates (top 1%)	*2019*
Shanghai Jiao Tong University Scholarship for Outstanding Undergraduates (ranked 1/58)	*2019*

OTHERS

Reviewer Service NeurIPS, ICLR, ICML, CVPR, ICCV, ECCV, RSS, RA-L.
Workshop Hoster CVPR 2026, IROS 2025. NeurIPS 2024, CVPR 2024, CVPR 2023.